

Producer consumer problem :

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define BUFFER_SIZE 5

int buffer[BUFFER_SIZE];
int in = 0;
int out = 0;

sem_t empty;
sem_t full;
pthread_mutex_t mutex;

void *producer(void *arg) {
    int item;
    for (int i = 0; i < 10; i++) {
        item = rand() % 100;
        sem_wait(&empty);
        pthread_mutex_lock(&mutex);

        buffer[in] = item;
        printf("Produced %d at %d\n", item, in);
        in = (in + 1) % BUFFER_SIZE;

        pthread_mutex_unlock(&mutex);
        sem_post(&full);
        sleep(1);
    }
    return NULL;
}

void *consumer(void *arg) {
    int item;
    for (int i = 0; i < 10; i++) {
        sem_wait(&full);
        pthread_mutex_lock(&mutex);

        item = buffer[out];
        printf("Consumed %d from %d\n", item, out);
        out = (out + 1) % BUFFER_SIZE;

        pthread_mutex_unlock(&mutex);
        sem_post(&empty);
        sleep(2);
    }
}
```

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    }
    return NULL;
}

int main() {
    pthread_t prod, cons;

    sem_init(&empty, 0, BUFFER_SIZE);
    sem_init(&full, 0, 0);
    pthread_mutex_init(&mutex, NULL);

    pthread_create(&prod, NULL, producer, NULL);
    pthread_create(&cons, NULL, consumer, NULL);

    pthread_join(prod, NULL);
    pthread_join(cons, NULL);

    sem_destroy(&empty);
    sem_destroy(&full);
    pthread_mutex_destroy(&mutex);

    return 0;
}

```

Output:

```

● wqadmin@DESKTOP-P0SNFU7:~/Desktop/code/os$ gcc prod_cons.c -o prod_cons -lpthread
● wqadmin@DESKTOP-P0SNFU7:~/Desktop/code/os$ ./prod_cons
Produced 83 at 0
Consumed 83 from 0
Produced 86 at 1
Produced 77 at 2
Consumed 86 from 1
Produced 15 at 3
Produced 93 at 4
Consumed 77 from 2
Produced 35 at 0
Produced 86 at 1
Consumed 15 from 3
Produced 92 at 2
Produced 49 at 3
Consumed 93 from 4
Produced 21 at 4
Consumed 35 from 0
Consumed 86 from 1
Consumed 92 from 2
Consumed 49 from 3
Consumed 21 from 4
○ wqadmin@DESKTOP-P0SNFU7:~/Desktop/code/os$

```